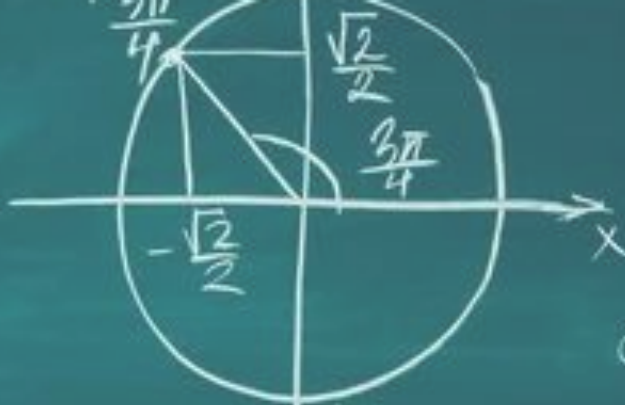
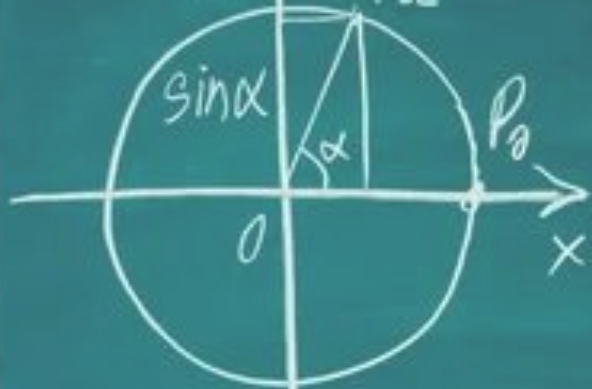




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0 \quad E(\sin) = E(\cos) = [1; 1]$$

First year Aerospace Engineering

Trigonometry exercises

Marcos Carrera Valle

Find the value of each function

- 1) $\sin(30^\circ)$
- 2) $\cos(\pi/4)$
- 3) $\tan(60^\circ)$
- 4) $\sin(\pi)$
- 5) $\cos(270^\circ)$
- 6) $\tan(0)$
- 7) $\cos(150^\circ)$
- 8) $\sin(4\pi/3)$
- 9) $\tan(315^\circ)$
- 10) $\sin(-\pi/6)$

Find the value of each function

- 11) $\sin(90^\circ) + \cos(180^\circ)$
- 12) $2 \times \sin(30^\circ)$
- 13) $\sin(45^\circ) \times \cos(45^\circ)$
- 14) $\tan(45^\circ) + \sin(270^\circ)$
- 15) $\cos(\pi/3) + \sin(\pi/6)$
- 16) $\tan(60^\circ) \times \tan(30^\circ)$
- 17) $\cos(0) - \sin(3\pi/2)$
- 18) $(\sin(60^\circ)) / (\cos(30^\circ))$
- 19) $\sin(\pi) \times \cos(\pi) + \tan(\pi/4)$
- 20) $\tan(225^\circ) - \cos(300^\circ)$

Find the Amplitude, Period, Frequency and Phase Shift for each sinusoidal wave

21) $y = 3\sin(4x)$

22) $y = 2\cos(\pi x - \pi)$

23) $y = 8\sin(x)\cos(x)$

24) $y = \cos^2(3x) - \sin^2(3x)$

25) $y = -\sin(2x + \pi/2)$

Solve the following equations in the domain $0 \leq x \leq 2\pi$

26) $\sin(2x) = \cos(x)$

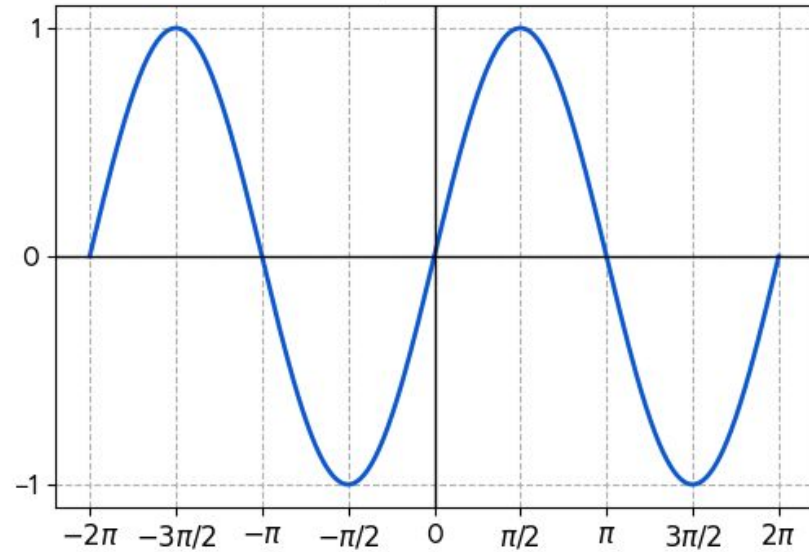
27) $\cos(2x) = -\cos(x)$

28) $\tan(2x) = -\tan(x)$

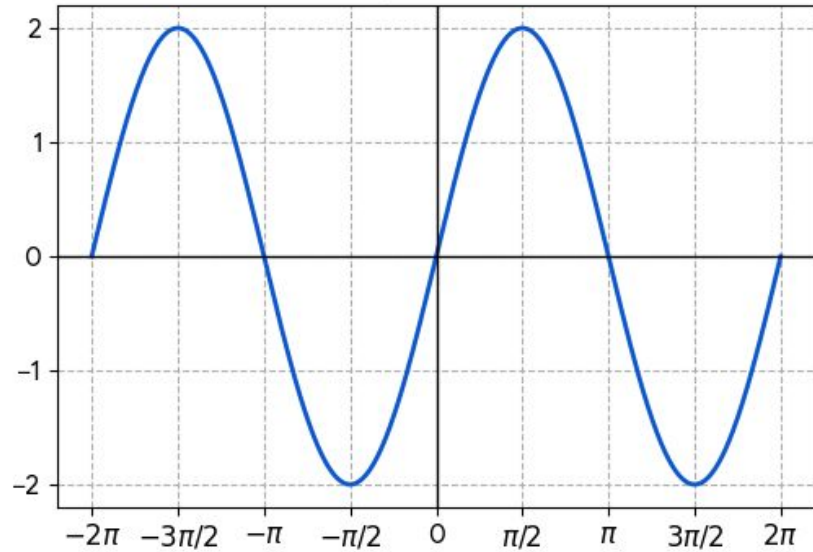
29) $\sin(x + \pi/6) = \cos(x)$

30) $2\cos^2(x) = 1 - \sin(x)$

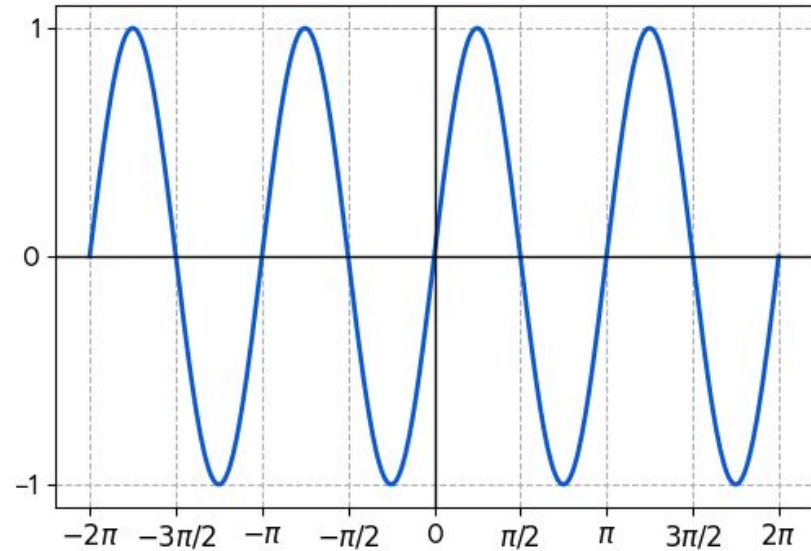
31) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



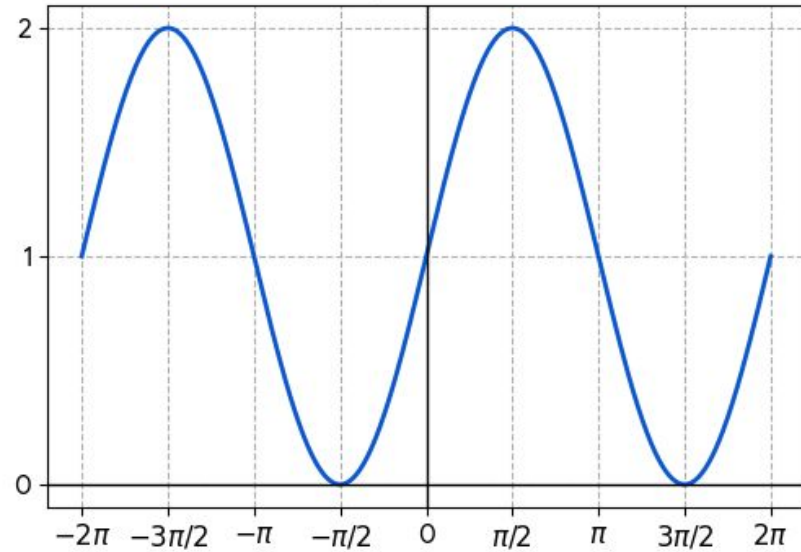
32) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



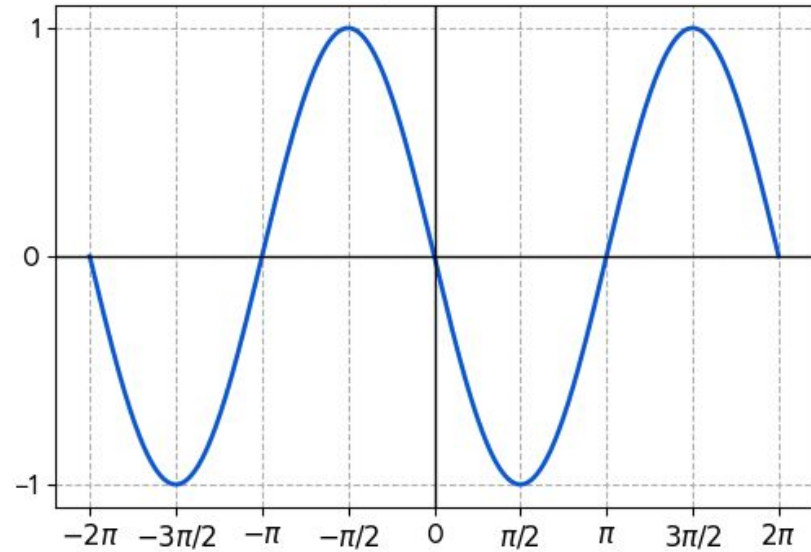
33) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



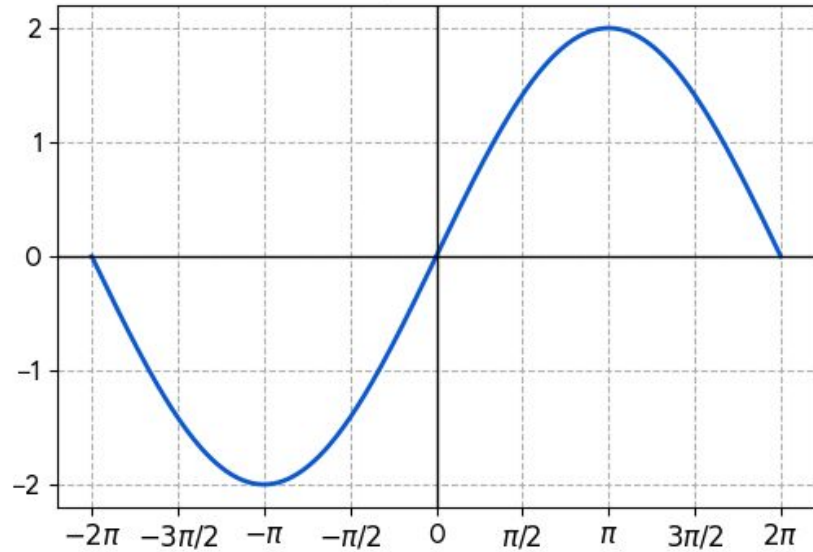
34) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



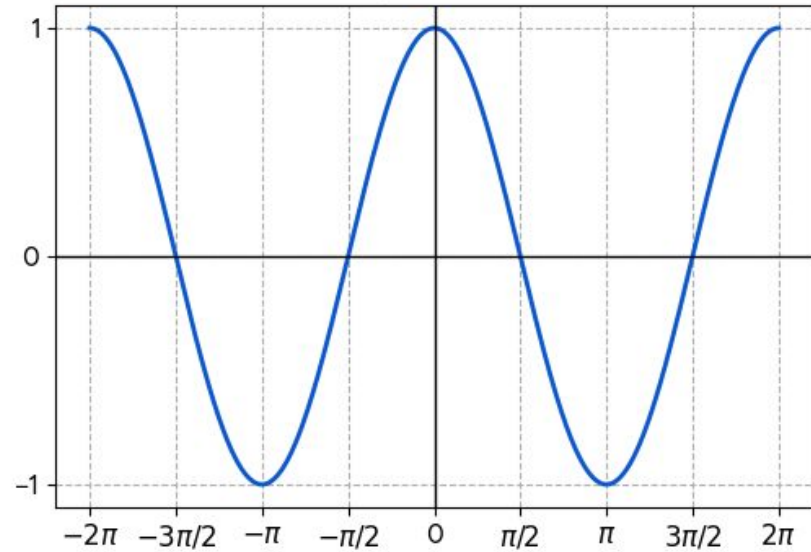
35) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



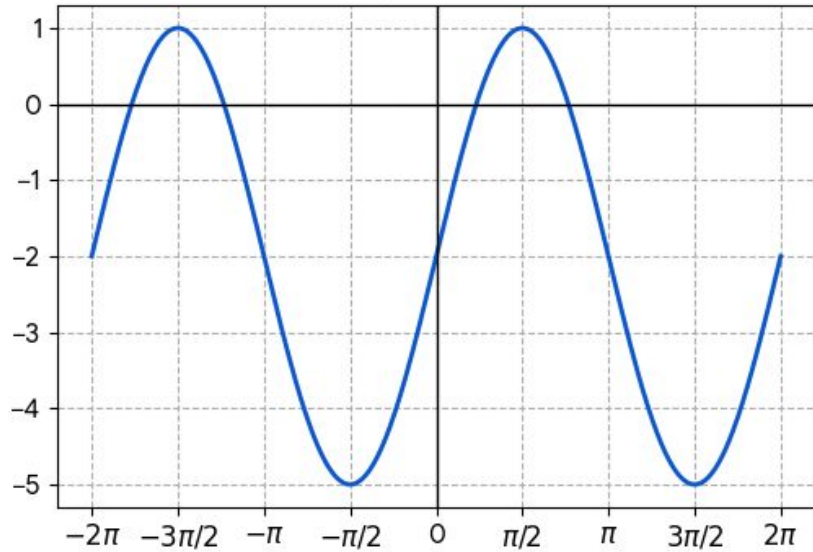
36) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



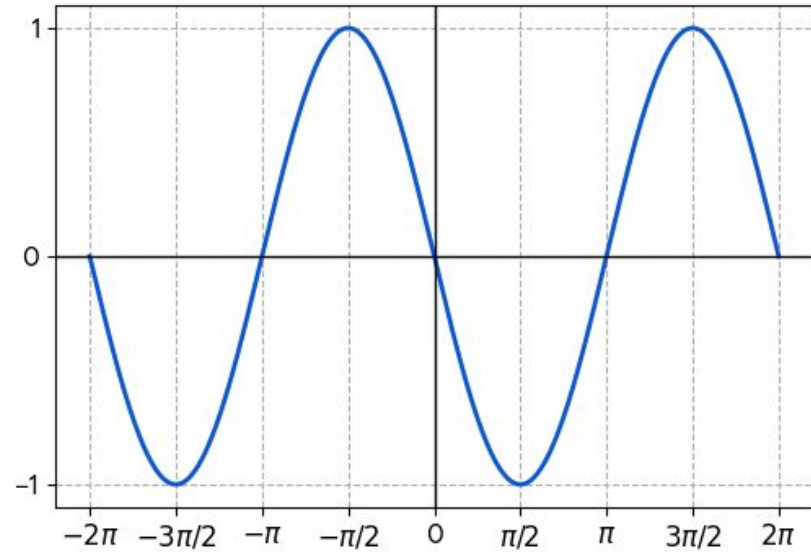
37) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



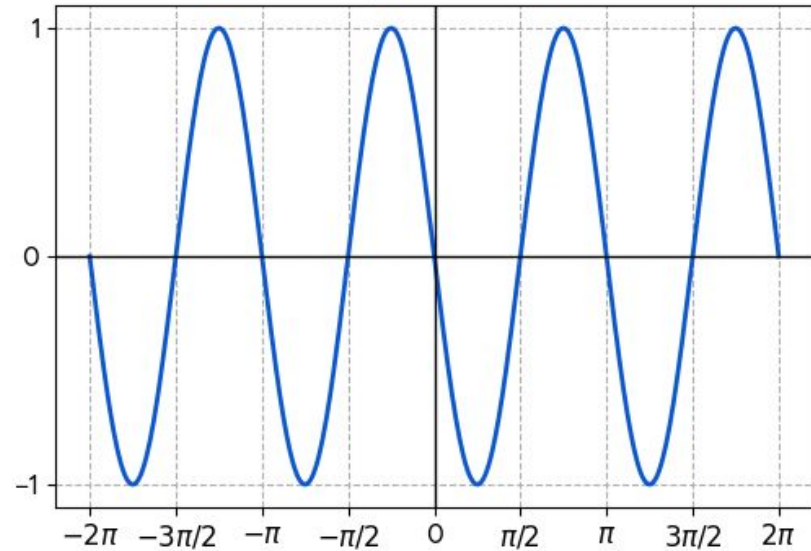
38) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



39) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



40) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$



Plot the following curves in the range $-2\pi \leq x \leq 2\pi$

41) $y = 2\sin(x)\cos(x)$

42) $y = \cos^2(x) - \sin^2(x)$

43) $y = (\sin(x) + \cos(x))^2$

44) $y = \sin(x + \pi/2)$

45) $y = \sin^2(x) + \cos^2(x) + \sin(x)$

46) $y = 1 - 2\sin^2(x/2)$

47) $y = \sin(2x) / \cos(x)$

48) $y = \cos(x)\cos(\pi) - \sin(x)\sin(\pi)$

49) $y = \cos^4(x) - \sin^4(x)$

50) $y = \sqrt{3}\sin(x) + \cos(x)$

SOLUTIONS

Find the value of each function

- 1) $1/2$
- 2) $\sqrt{2}/2$
- 3) $\sqrt{3}$
- 4) 0
- 5) 0
- 6) 0
- 7) $-\sqrt{3}/2$
- 8) $-\sqrt{3}/2$
- 9) -1
- 10) $-1/2$

Find the value of each function

- 11) 0
- 12) 1
- 13) $1/2$
- 14) 0
- 15) 1
- 16) 1
- 17) 2
- 18) 1
- 19) 1
- 20) $1/2$

Find the Amplitude, Period, Frequency and Phase Shift for each sinusoidal wave

- 21) Amplitude: 3, Period: $\pi/2$, Frequency: $2/\pi$, Phase Shift: 0
- 22) Amplitude: 2, Period: 2, Frequency: $1/2$, Phase Shift: 1 (to the right)
- 23) Amplitude: 4, Period: π , Frequency: $1/\pi$, Phase Shift: 0
- 24) Amplitude: 1, Period: $\pi/3$, Frequency: $3/\pi$, Phase Shift: 0
- 25) Amplitude: 1, Period: π , Frequency: $1/\pi$, Phase Shift: $-\pi/4$ (to the left)

Solve the following equations in the domain $0 \leq x \leq 2\pi$

26) $\pi/6, \pi/2, 5\pi/6, 3\pi/2$

27) $\pi/3, \pi, 5\pi/3$

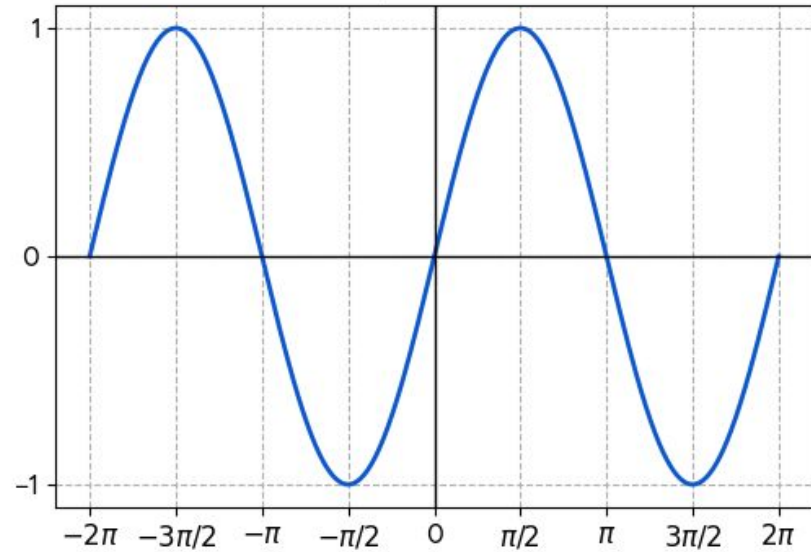
28) $0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3$

29) $\pi/6, 7\pi/6$

30) $\pi/2, 7\pi/6, 11\pi/6$

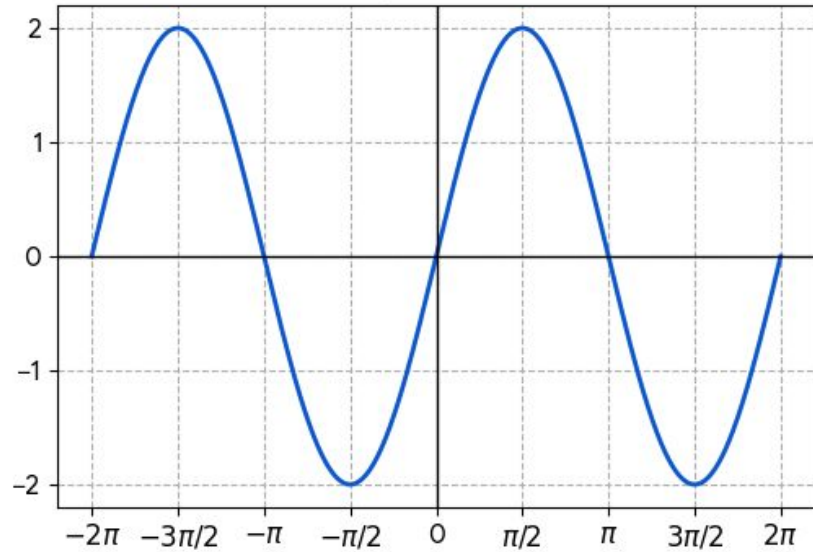
31) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = \sin(x)$$



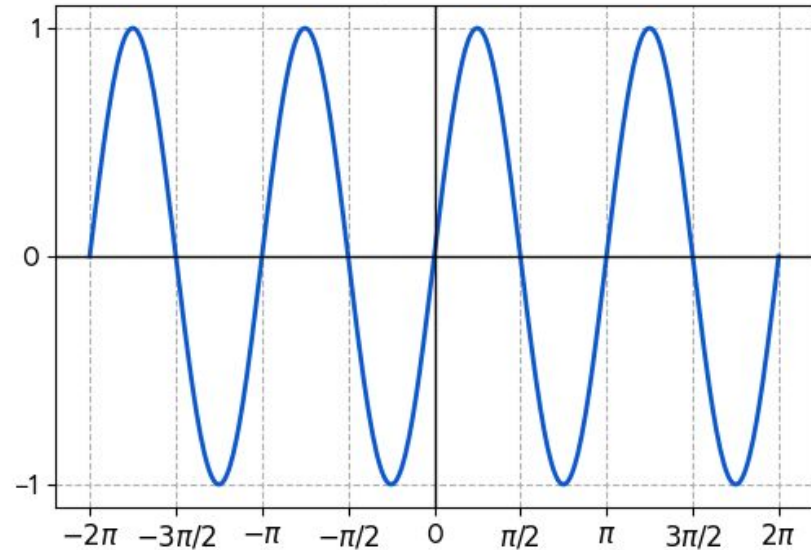
32) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = 2\sin(x)$$



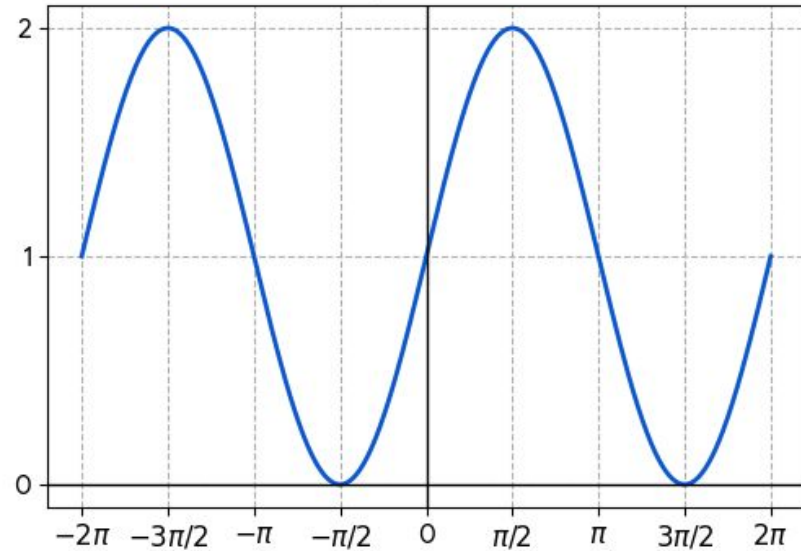
33) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = \sin(2x)$$



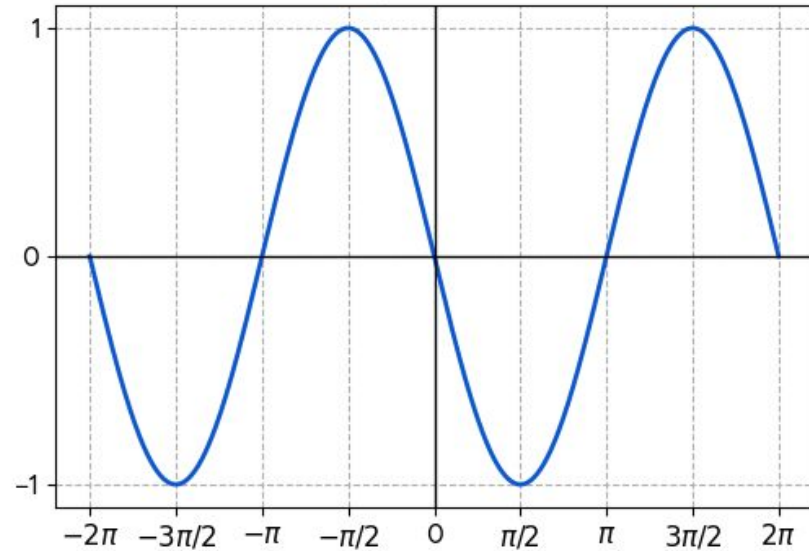
34) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = 1 + \sin(x)$$



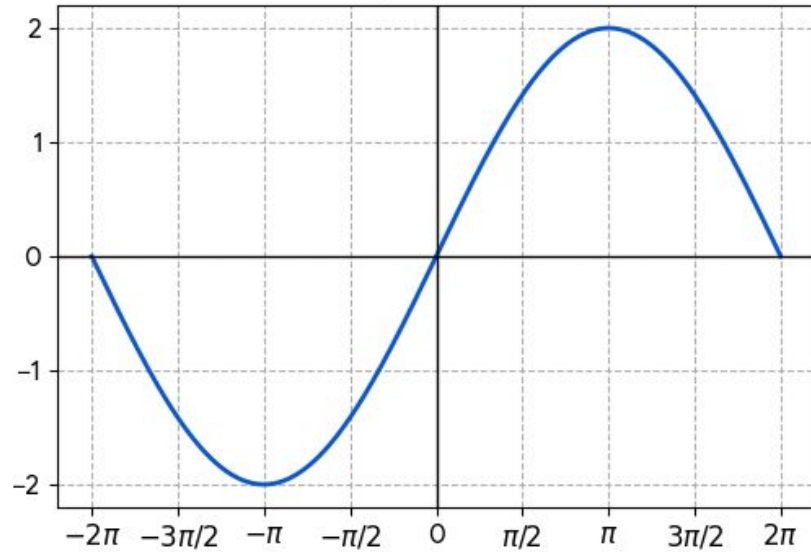
35) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = \sin(x - \pi)$$



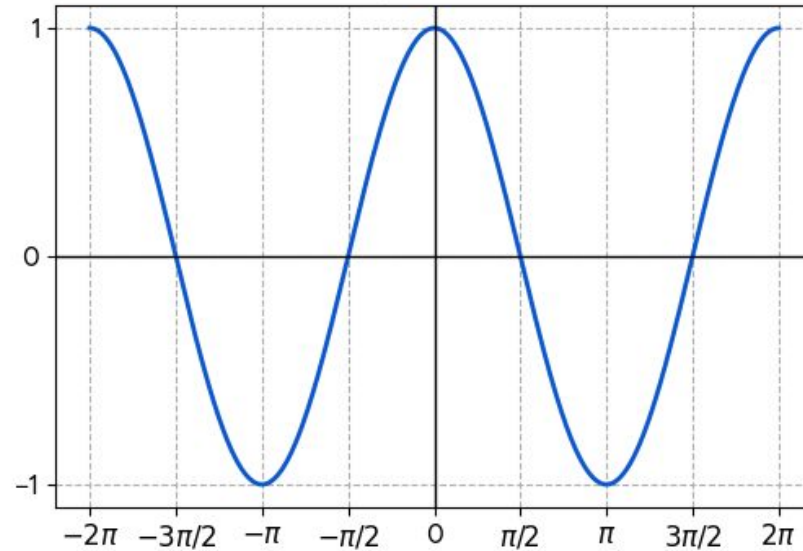
36) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = 2\sin(0.5x)$$



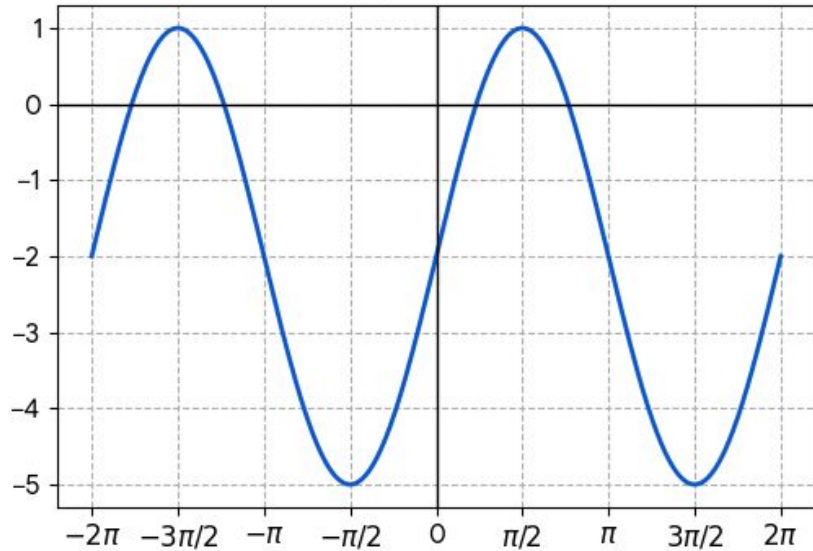
37) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = \sin(x + \pi/2)$$



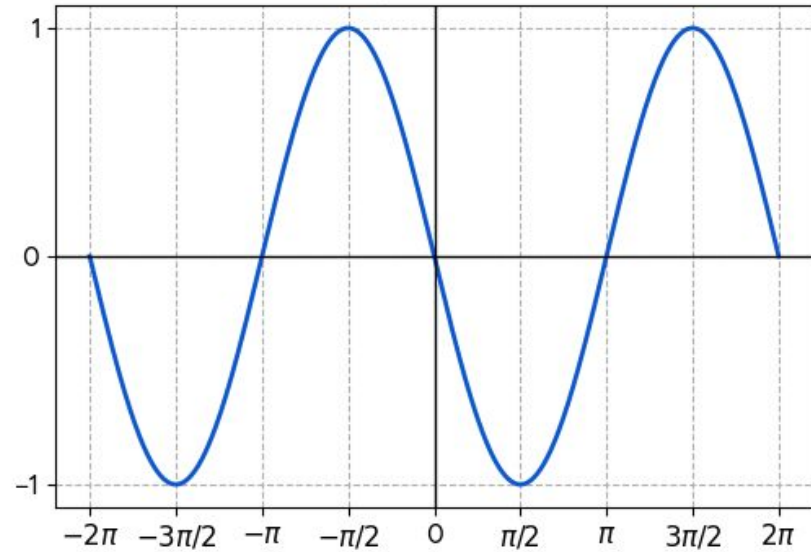
38) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = -2 + 3\sin(x)$$



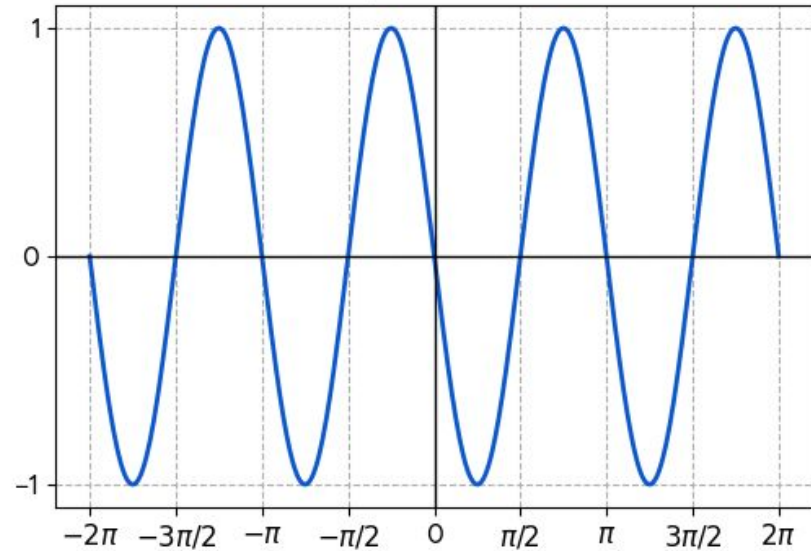
39) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = -\sin(x)$$



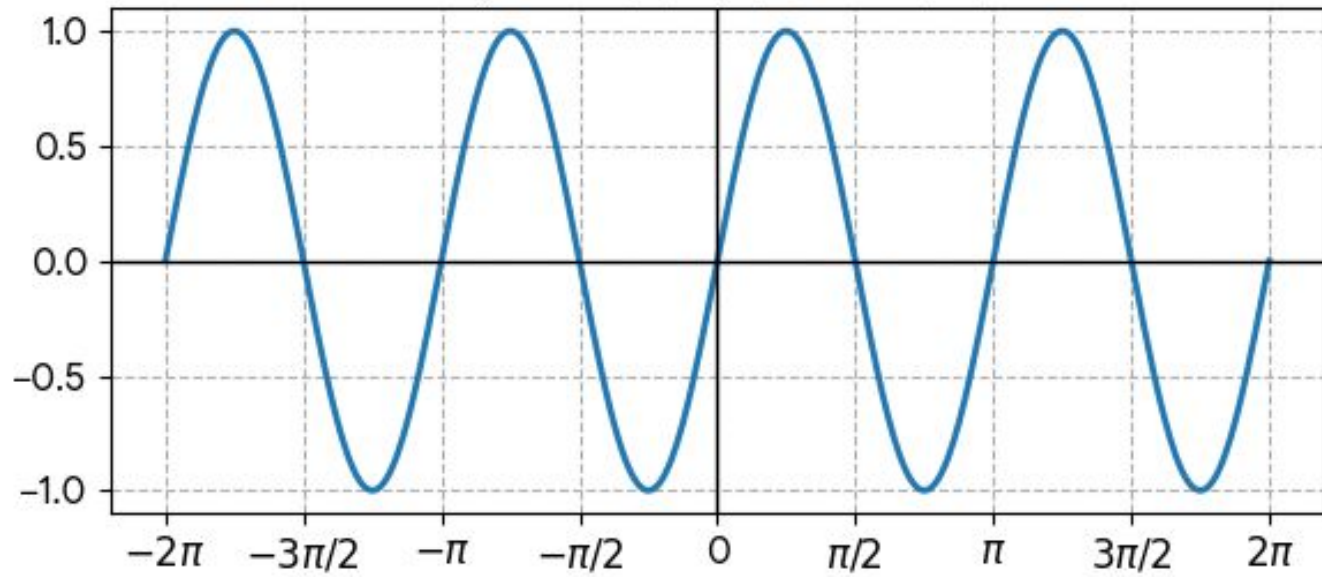
40) Find the function represented by this curve in the form $f(x) = a + b \cdot \sin(c \cdot x - d)$

$$f(x) = \sin(2x - \pi)$$

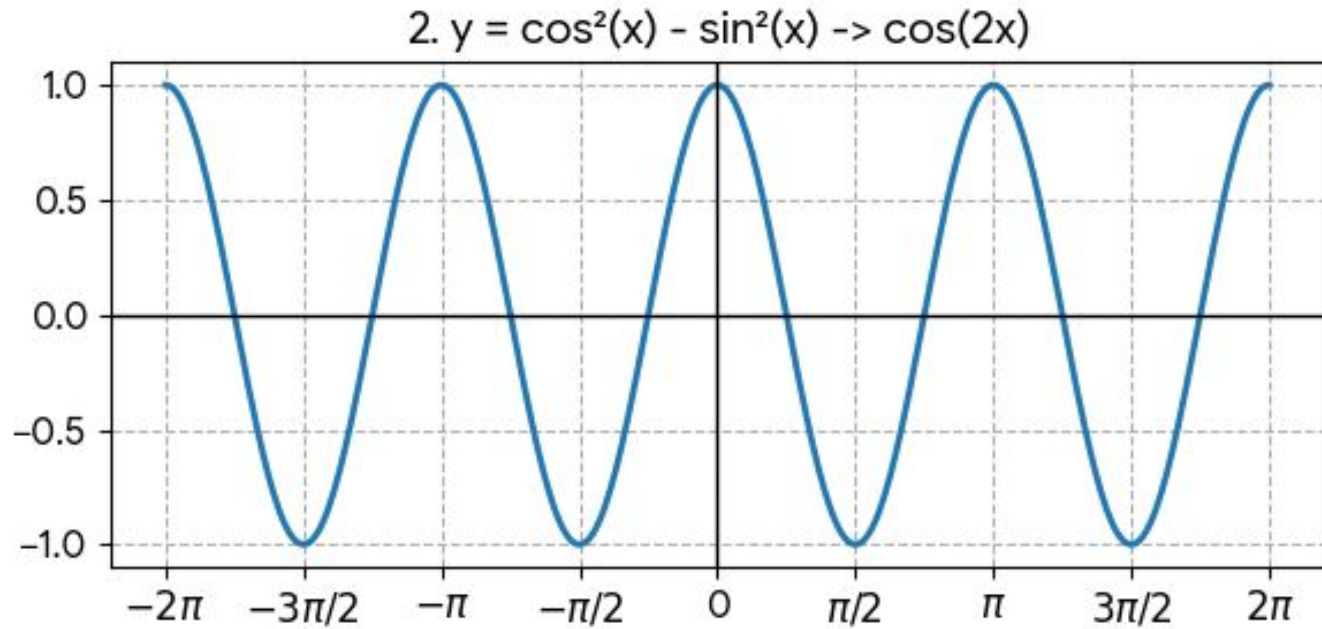


41-50)

1. $y = 2\sin(x)\cos(x) \rightarrow \sin(2x)$

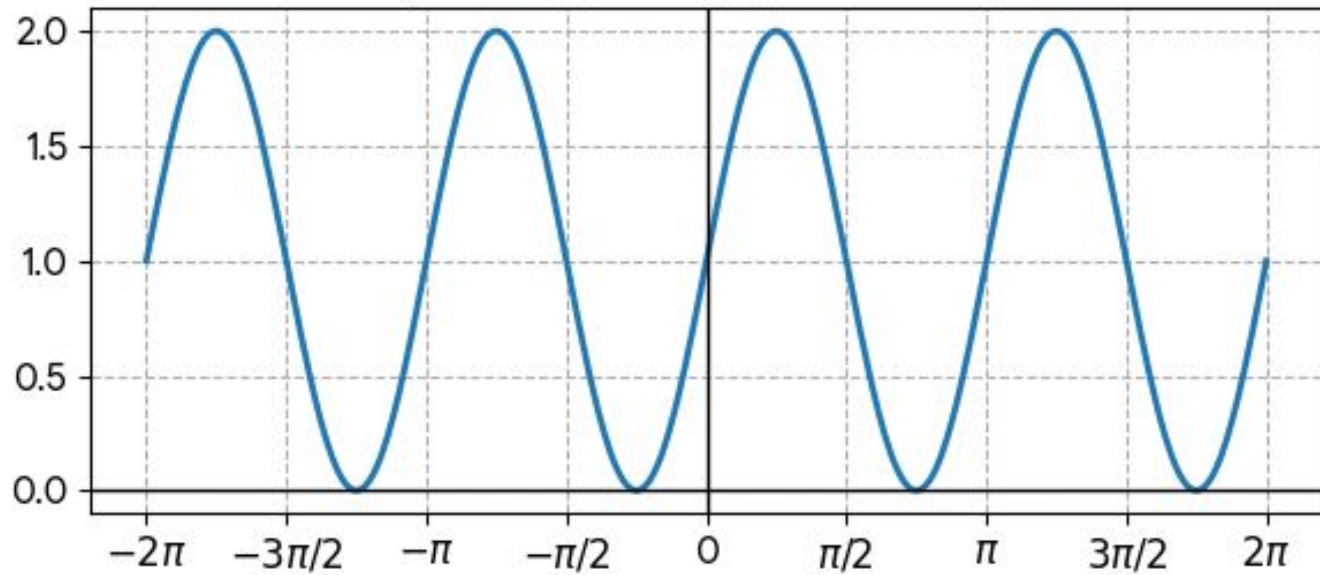


41-50)



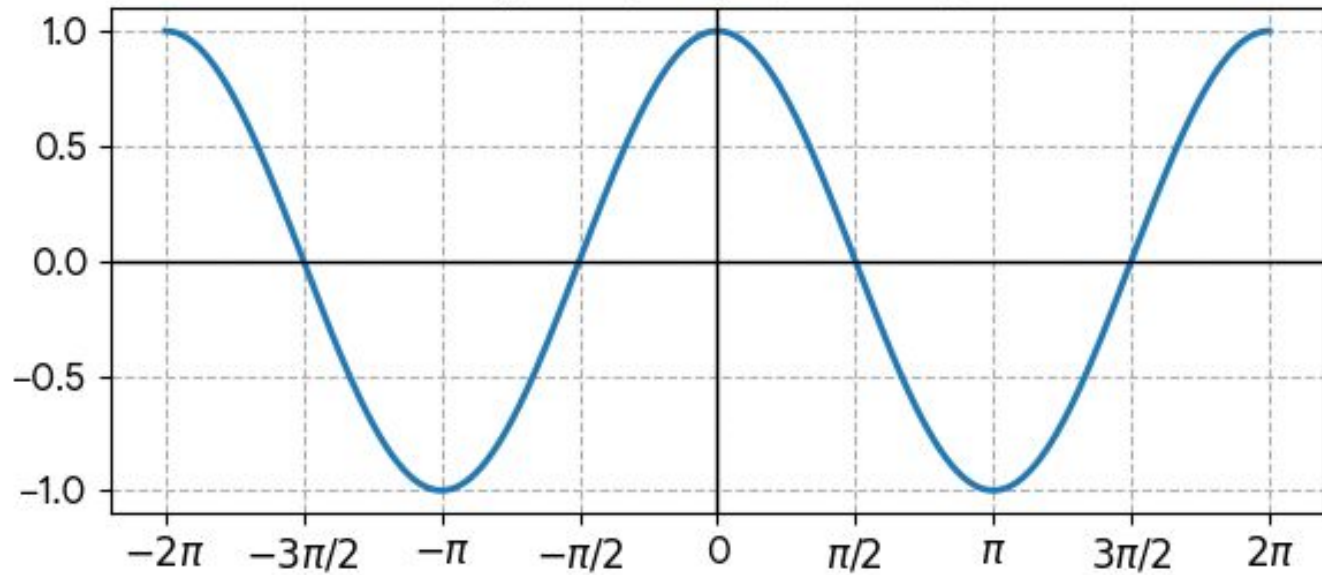
41-50)

3. $y = (\sin(x) + \cos(x))^2 \rightarrow 1 + \sin(2x)$



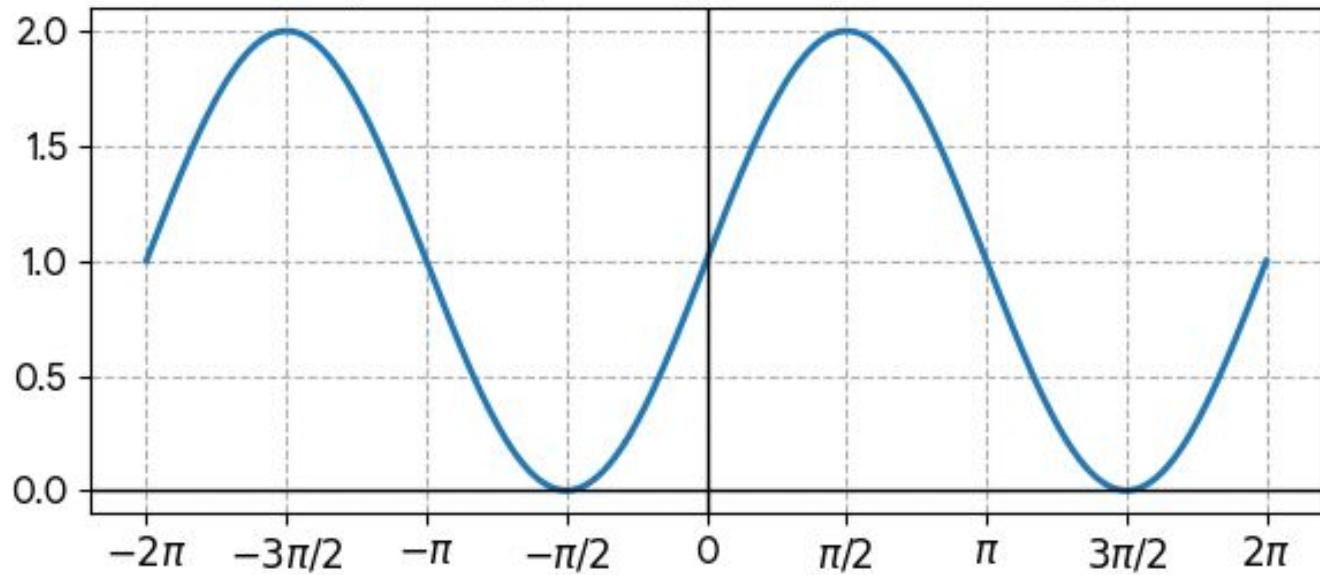
41-50)

4. $y = \sin(x + \pi/2) \rightarrow \cos(x)$



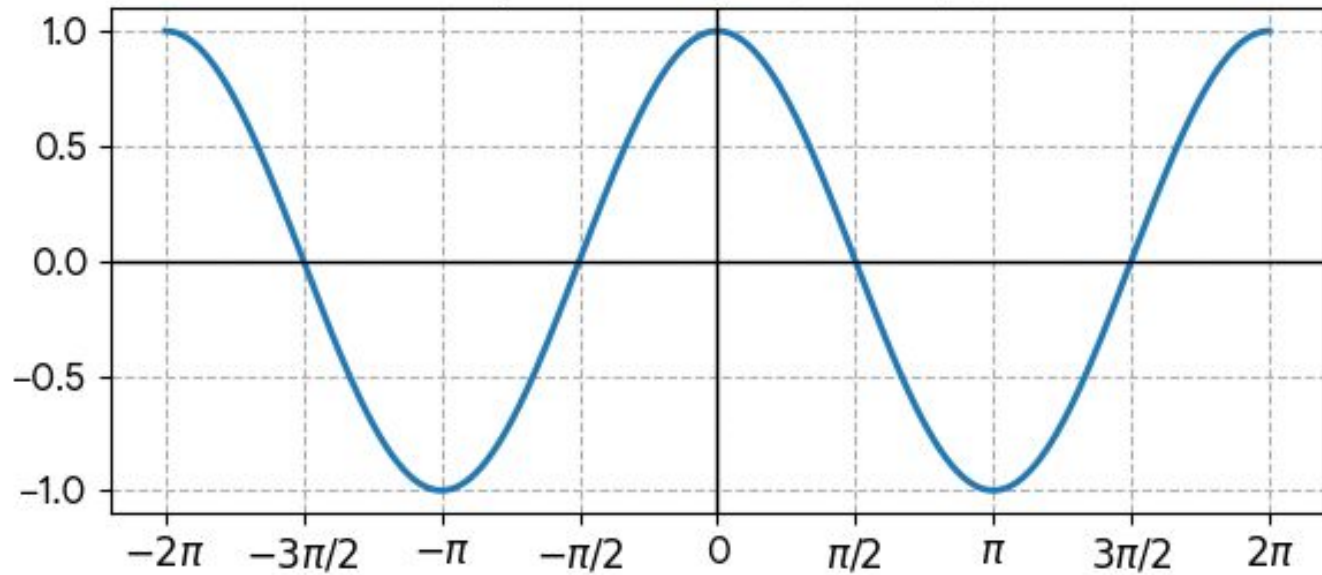
41-50)

5. $y = \sin^2(x) + \cos^2(x) + \sin(x) \rightarrow 1 + \sin(x)$



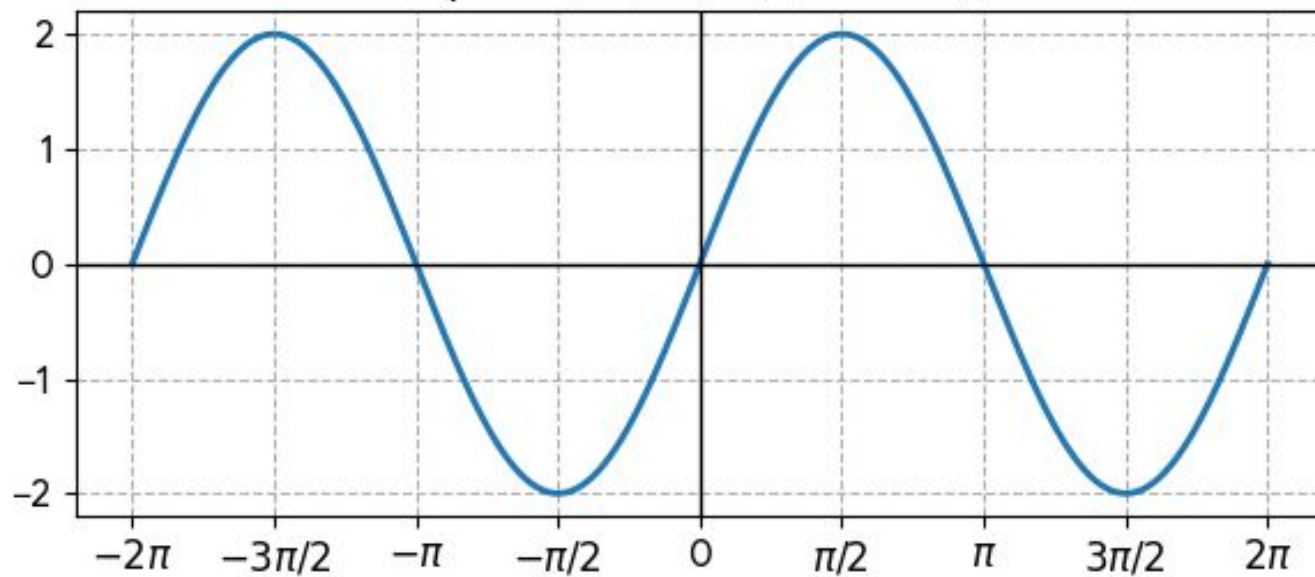
41-50)

6. $y = 1 - 2\sin^2(x/2) \rightarrow \cos(x)$



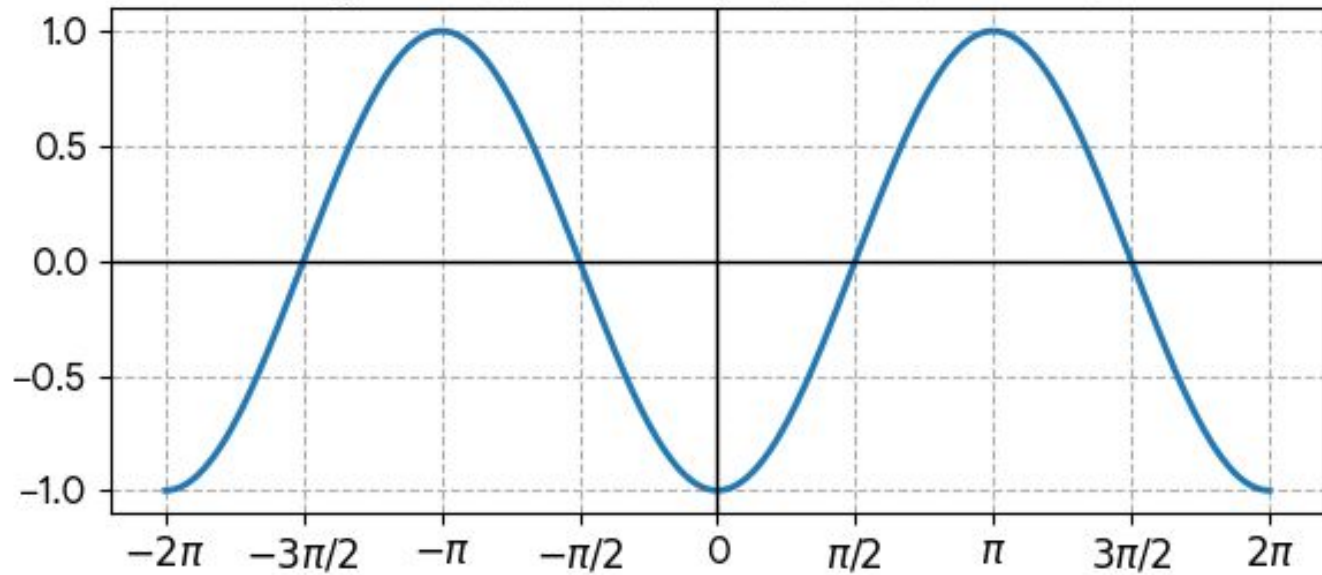
41-50)

7. $y = \sin(2x) / \cos(x) \rightarrow 2\sin(x)$

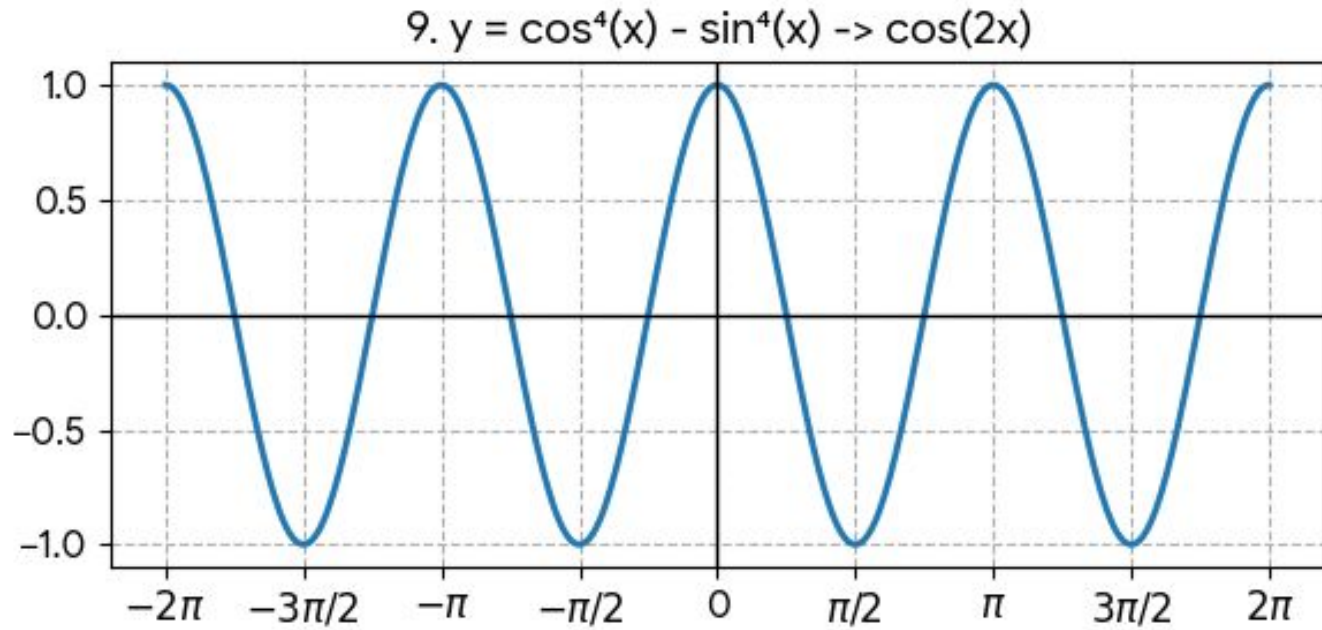


41-50)

8. $y = \cos(x)\cos(\pi) - \sin(x)\sin(\pi) \rightarrow -\cos(x)$



41-50)



41-50)

10. $y = \sqrt{3}\sin(x) + \cos(x) \rightarrow 2\sin(x + \pi/6)$

